Power11
Quantum Safety

Cryptography touches every corner of the digital world

Internet Protocols

Hyper-text Transfer Protocol Secure (HTTPS), Transport Layer Security (TLS), Domain Name Service(DNS), SFTP, …

Critical Infrastructure

Code updates; Industrial Control systems, Gas pipelines, smart grids, transportation systems …

Blockchain Applications

Coin wallets, Transactions, Authentication

Digital Signature Laws

EiDAS - PDF Advanced Electronic Signature – (PAdES), Advanced Electronic Signatures (AES), …

Financial Systems

Payment Systems: (EMV, SWIFT, Settlement Systems, FinTech, …)

Enterprise Applications

EMAIL – PGP, Identity Management PKI/LDAP/..., PKI Services

Cryptography is used in trillions of transactions on billions of devices

Internet
- Domain name system (DNS)
- Hypertext Transfer Protocol Secure (HTTPS)
- SSH File Transfer Protocol (FTP)

Digital signatures
- Electronic identification and trust services (eIDAS)
- PDF advanced electronic signature (PAdES)
- Advanced electronic signatures

Critical infrastructure
- Code updates
- Control systems
- Car systems

Financial systems
- Payment systems

Enterprise
- Email
- Identity management
- LDAP
- PKI services
- Bespoke applications

Documents that needs to stay secure for a long period of time

Passports: 10 years from issue

Road vehicles: 15–20 years

Aircraft/rail: 25–30 years

Some critical infrastructure: 50+ years

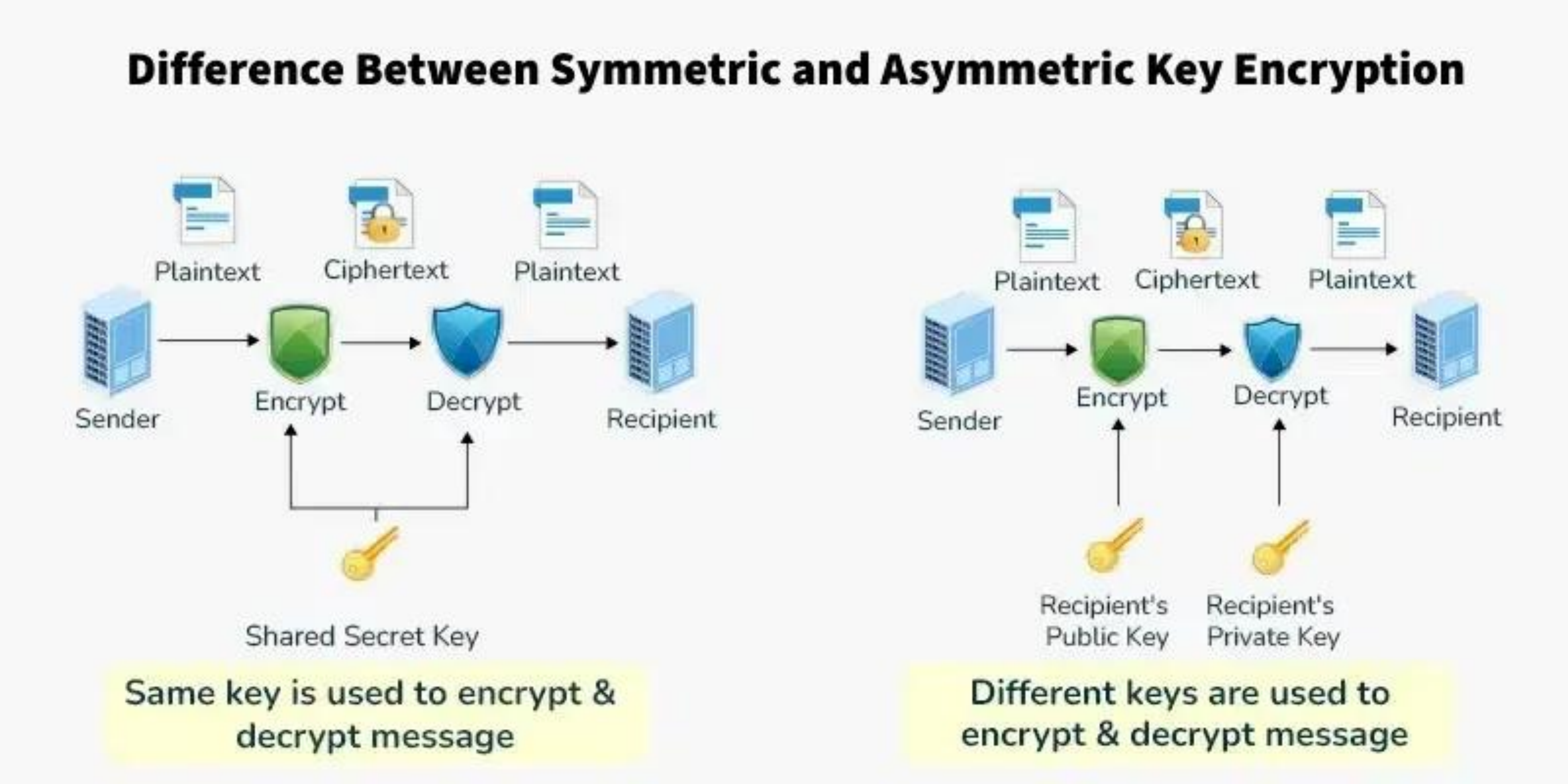
Data needs to stay secure for a long time

HIPAA: 6 years from last use per Security Rule

Tax records: 7–10 years in most countries; Sarbanes-Oxley Act set the precedent in the US

Legitimate interest under GDPR: 20+ years

Difference between Symmetric and Asymmetric Key Encryption



Currently considered quantum safe

Can be attacked with quantum computers

Content taken from https://www.geeksforgeeks.org/computer-networks/difference-between-symmetric-and-asymmetric-key-encryption/

Why Quantum Computers will be able to break existing asymmetric encryption algorithms

Much of today’s cryptography relies on hard mathematical problems.

The most powerful classical computer today would take millions of years to find the solution.

However, Quantum Computers will be able to break this problem within hours.

Definitions

Factor

Numbers you can multiply to get original number

Factors of **12**:
1 2 3 4 6 12

Factors of **7**:
1 7

Prime

Numbers whose factors are only 1 and itself

2 3 5 7 11
13 29 37 61

Number divisible by only 1 and itself

Semi-Prime

Numbers whose factors are prime numbers

Factors of Semi Prime **21**:
1 **3** **7** 21

Product of two Primes is always Semi Prime

Modulo

Remainder Division

13 MOD 5 = 3

21 MOD 5 = 1

25 MOD 5 = 0

Existing encryption algorithms

Why Quantum Computers will be able to break them

Security of encryption algorithms lies in the difficulty of factoring Semi Prime numbers.

- If given the number **133**, could you extract **7** and **19**?
- Really, how about **1,909**? (23×83) How about **55,958,993**? (7001×7993)

Only 11 Bits

Only 26 Bits

In 1991, the RSA Laboratory created the RSA Challenge

- Released 54 Semi-Primes of various sizes and asked for Factors (with cash prices)
  - Competition ended in 2007 – only 12 Semi-Primes were successfully factored
  - As of 2020, another 11 were identified (no more cash prices were awarded)
    - Biggest number factored: 829 Bits (February 2020)
    - In 34 years, the 1024 bit number has never been factored

1024 bit RSA keys was the recommended standard since 2002
2048 bit RSA keys is the recommended standard since 2015

Largest factorized RSA number (829 bits) so far...

RSA-250

- RSA-250 has 250 decimal digits (829 bits), and was factored in February 2020 by Fabrice Boudot, Pierrick Gaudry, Aurore Guillevic, Nadia Heninger, Emmanuel Thomé, and Paul Zimmermann. The announcement of the factorization occurred on February 28, 2020.

RSA-250 = 2140324650240744961264423072839333563008614715144755017797754920881418023447
140136643345519095804679610992851872470914587687396261921557363047454770520805119056493106687691590019759405693457452230589325976697471681738069364894699871578494975937497937

RSA-250 = 6413528947707158027879019017057738908482501474294344720811685963202453234463
0238623598752668347708737661925585694639798853367
× 3337202759497815655622601060535511422794076034476755466678452098702384172921
0037080257448673296881877565718986258036932062711

The factorisation of RSA-250 utilized approximately **2700 CPU core-years**, using a 2.1 GHz Intel Xeon Gold 6130 CPU as a reference. The computation was performed with the Number Field Sieve algorithm, using the open source CADO-NFS software.

The team dedicated the computation to Peter Montgomery, an American mathematician known for his contributions to computational number theory and cryptography who died on February 18, 2020, and had contributed to factoring RSA-768.

Content taken from https://en.wikipedia.org/wiki/RSA_numbers

1,024 bits and 2,048 bits RSA Challenge numbers

RSA-1024 (1,024 bits)

- RSA-1024 has 309 decimal digits and has not been factored so far. $100,000 was previously offered for factorization.

RSA-1024 = 135066410865995223349603216278805969938881475605667027524485143851526510604
859533833940287150571909441798207282164471551373680419703964191743046496589
274256239341020864383202110372958725762358509643110564073501508187510676594
629205563685529475213500852879416377328533906109750544334999811150056977236
890927563

RSA-2048 (2,048 bits)

- RSA-2048 has 617 decimal digits. It is the largest of the RSA numbers and carried the largest cash prize for its factorization, $200,000.

RSA-2048 = 251959084756578934940271832400483985714292821262040320277771378360436620207
075955562640185258807844069182906412495150821892985591491761845028084891200728
449926873928072877767359714183472702618963750149718246911650776133798590957
000973304597488084284017974291006424586918171951187461215151726546322822168699
875491824224336372590851418654620435767984233871847744479207399342365848238
242811981638150106748104516603773060562016196762561338441436038339044149526
344321901146575444541784240209246165157233507787077498171257724679629263863563
732899121548314381678998850404453640235273819513786365643912120103971228221
20720357

Content taken from https://en.wikipedia.org/wiki/RSA_numbers

Shor's algorithm

Shor's algorithm is a quantum algorithm for finding the prime factors of an integer. It was developed in 1994 by the American mathematician Peter Shor.[1][2] It is one of the few known quantum algorithms with compelling potential applications and strong evidence of superpolynomial speedup compared to best known classical (non-quantum) algorithms.[3] However, beating classical computers will require millions of qubits due to the overhead caused by quantum error correction.[4]

Feasibility and impact

If a quantum computer with a sufficient number of qubits could operate without succumbing to quantum noise and other quantum-decoherence phenomena, then Shor's algorithm could be used to break public-key cryptography schemes, such as

- The RSA scheme
- The finite-field Diffie–Hellman key exchange
- The elliptic-curve Diffie–Hellman key exchange[9]

RSA can be broken if factoring large integers is computationally feasible. As far as is known, this is not possible using classical (non-quantum) computers; no classical algorithm is known that can factor integers in polynomial time. However, Shor's algorithm shows that factoring integers is efficient on an ideal quantum computer, so it may be feasible to defeat RSA by constructing a large quantum computer. It was also a powerful motivator for the design and construction of quantum computers, and for the study of new quantum-computer algorithms. It has also facilitated research on new cryptosystems that are secure from quantum computers, collectively called post-quantum cryptography.

Quantum Computers are bringing a new challenge

Quantum Computers...

IBM Quantum System 1

IBM Quantum System 2

...are able to solve some classes of problems orders of magnitude faster than classical computers

... and at the same time able to break most of the world's existing cryptographic algorithms "Quantum Attacks"

- Any system that uses current algorithms is at risk of exposing (customer) data
- According to various reports this will take place around 2030.
  - ➢ See NIST report and IBM Quantum Safe Cryptography for details

AND, *cybercriminals are not waiting for Quantum Computers*

Harvest now, decrypt later

Now

Availability of "cryptographically relevant" quantum computers

Later

Harvest confidential data to decrypt later

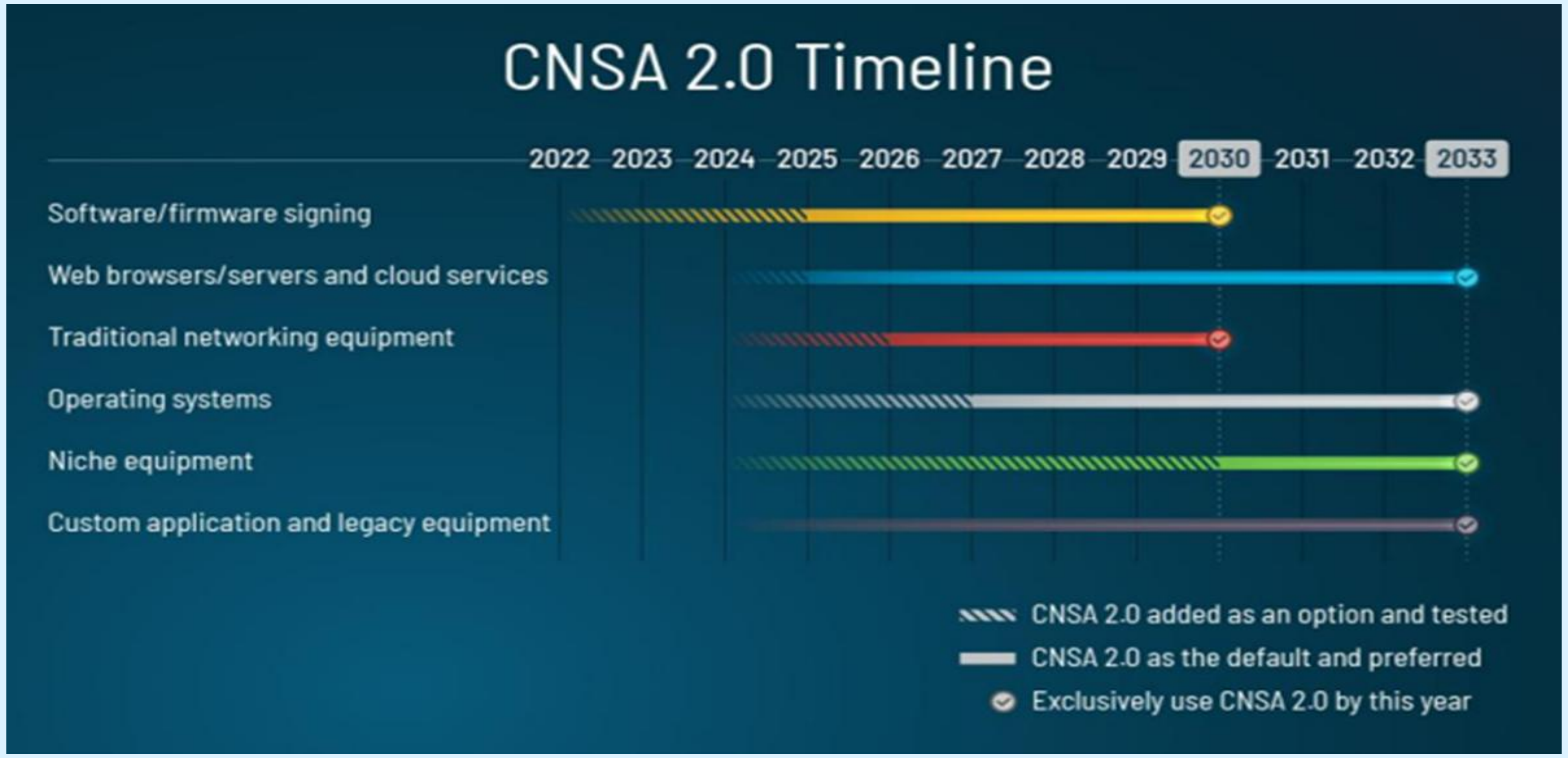
Decrypt lost or harvested confidential data by breaking encryption

Disrupt business with manipulation through fraudulent authentication

Manipulate digitally signed contracts and legal history by forging digital signatures

Government Mandates puts heightened focus on Quantum Safe

US Government mandate for quantum-safe federal agencies



CNSA 2.0: Quantum-safe standards are preferred for national security systems by the mid-2020s and required by the early 2030s to defend against quantum computing threats.

Transitioning to quantum-safe communication:
Adding Q-safe preference to OpenSSL TLSv1.3

- OpenSSL v3.5 enables TLS clients to send multiple key shares and introduces a new group selection algorithm for servers to address the NSA's 'prefer' quantum-safe algorithms mandate in CNSA 2.0.
- Future quantum computers will be able to break the asymmetric cryptographic algorithms widely in use online today.
- The US National Security Agency has issued a list of approved quantum-safe (or Q-safe) algorithms, as well as a timeline for transition in the CNSA 2.0, Commercial National Security Algorithm Suite.
- For TLS (Transport Layer Security, RFC 8446), the approved algorithms for key agreements and certificates are ML-KEM (FIPS-203) and ML-DSA (FIPS-204) or SPINCS+ (FIPS-205) respectively.
- These were all co-developed by researchers at IBM.

It is expected that all major Linux distributions will soon include OpenSSL v3.5!

Content taken from https://research.ibm.com/blog/openssl-quantum-safe-cnsa

Quantum Safe firmware signing

Power11 Hybrid Secure Boot: *Protects integrity of platform against quantum attacks*

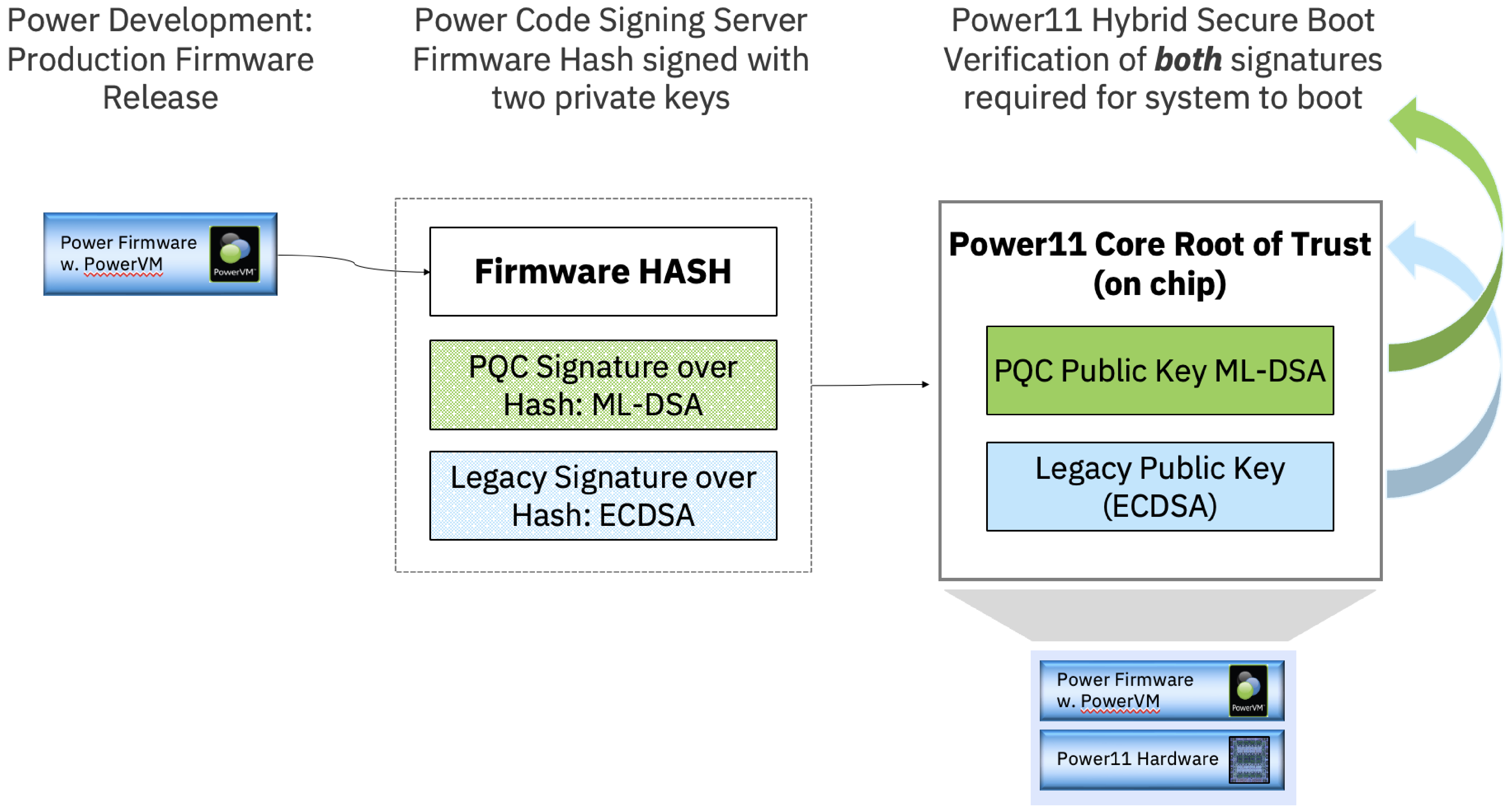
How?

if…

…an attacker *generates a malicious version of Power firmware*

…and if the attacker compromise the legacy signature via a Quantum attack, legacy signature verification will succeed

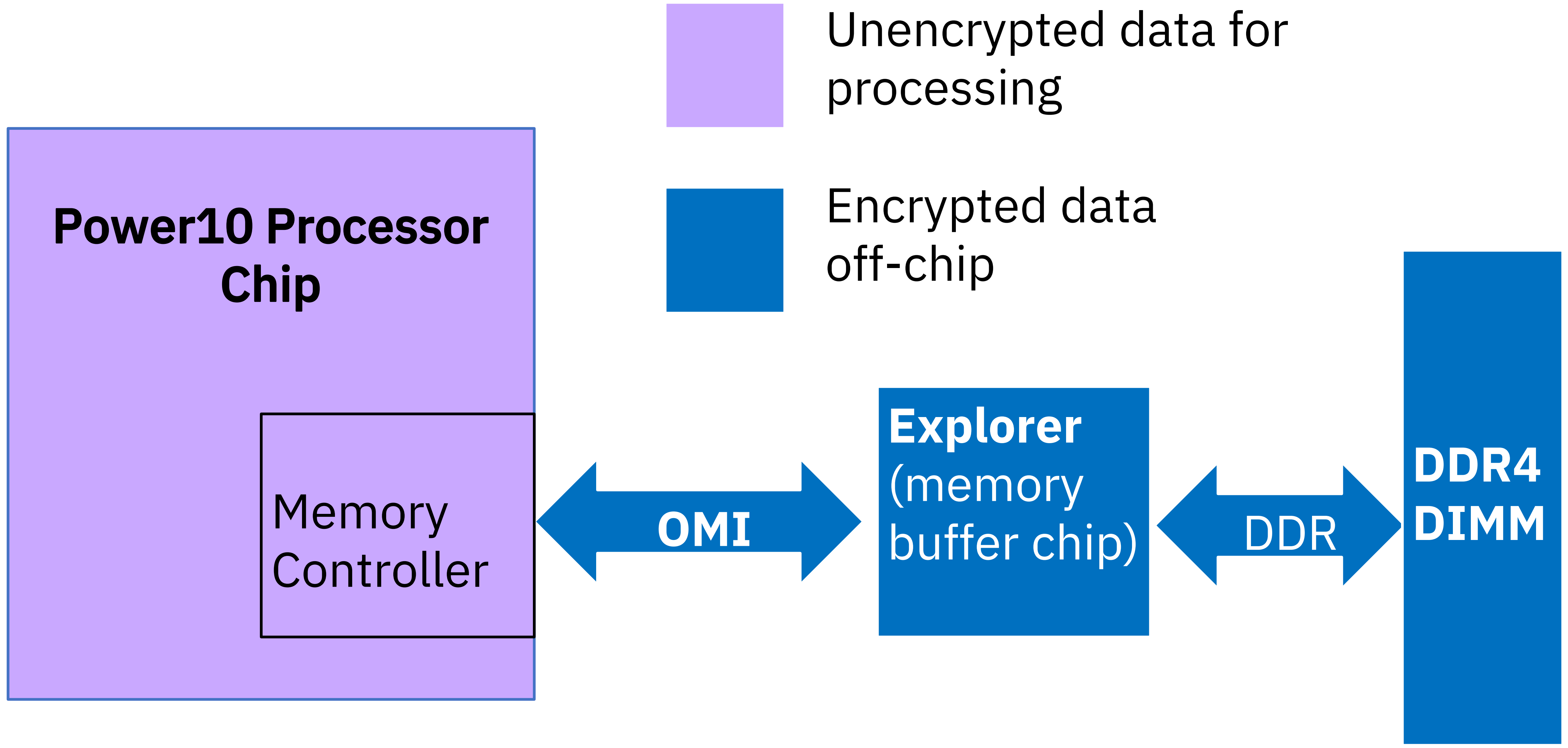
However, *PQC signature verification will still fail, preventing the system from booting the malicious firmware*



Transparent Memory Encryption

Power10 Processor provides support for encryption of main memory off chip

No additional setup in OS or application: firmware controlled
No CPU cycles spent, no impact on memory throughput

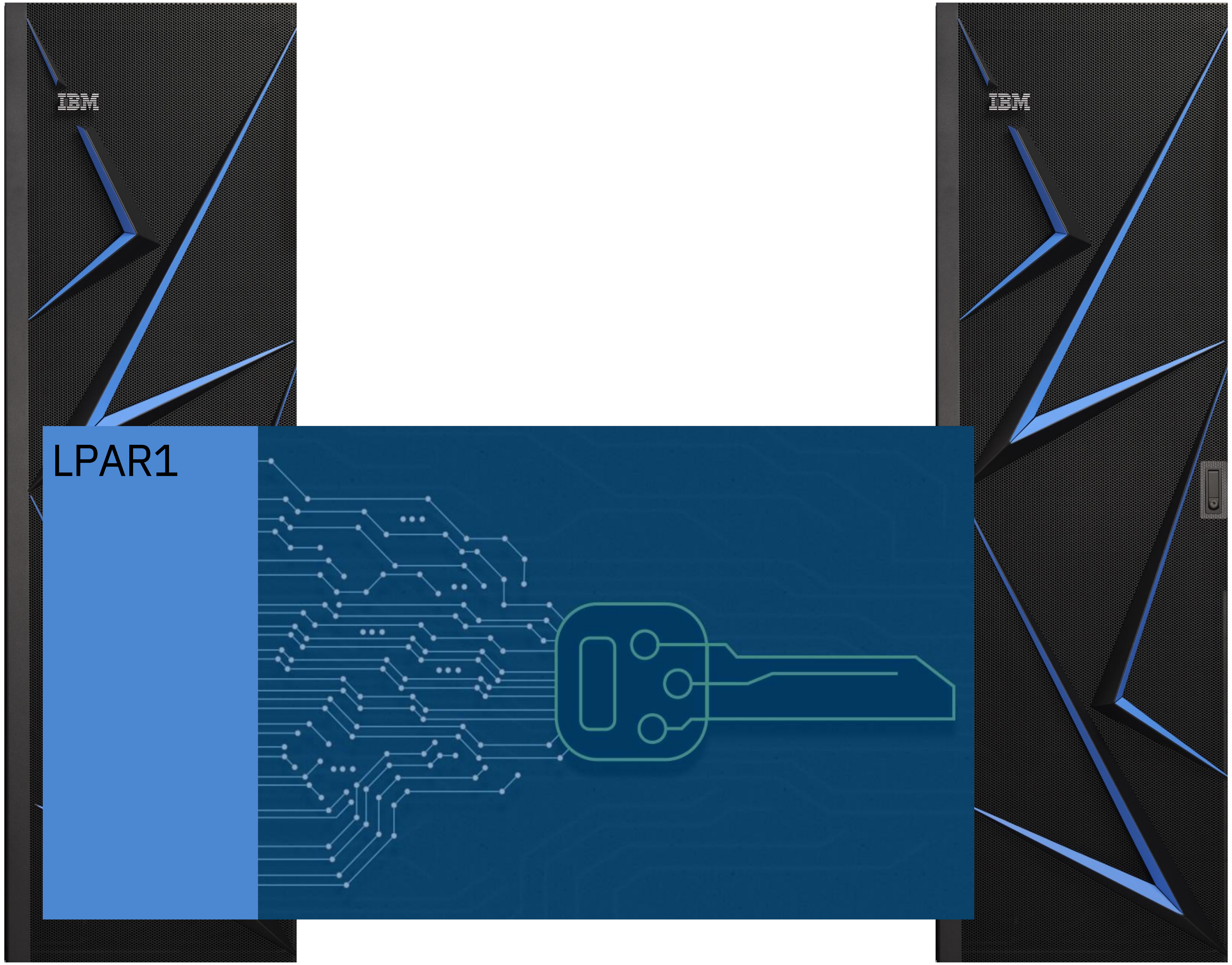


Protection against

- Bus probing attacks
- Cold boot attacks
- Future proof for adoption of non-volatile memory technologies for main memory
- Quantum safe because of symmetric encryption method

Live Partition Mobility

- Partitions are encrypted if/when they need to be migrated from one system to another
- Protects the data in motion
- Quantum safe because of symmetric encryption method



Quantum Safe Live Partition Mobility (LPM)

Power11 Encrypted LPM: *Protects from “capture now, decrypt later” attacks.*

How?

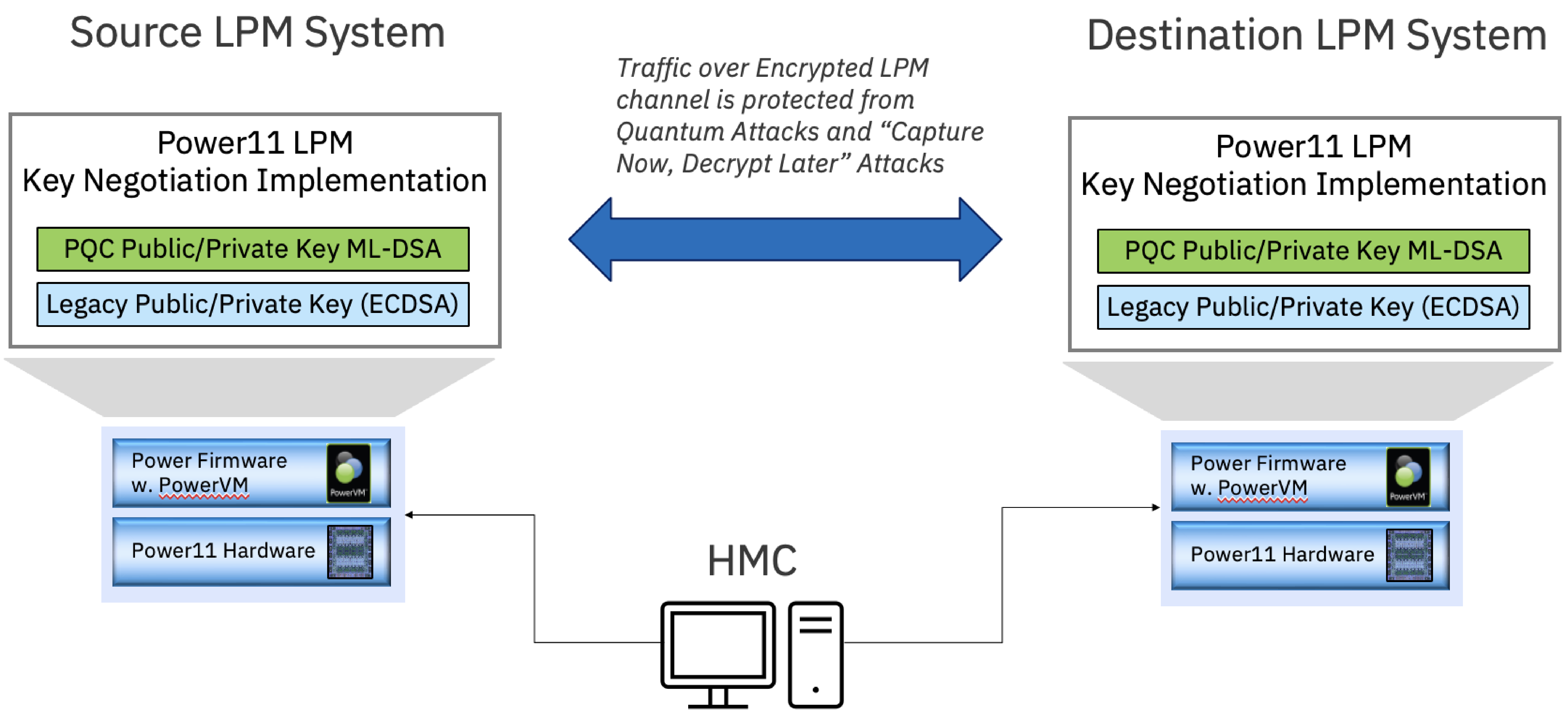
2 public/private key pairs are used in the LPM key negotiation.

if…

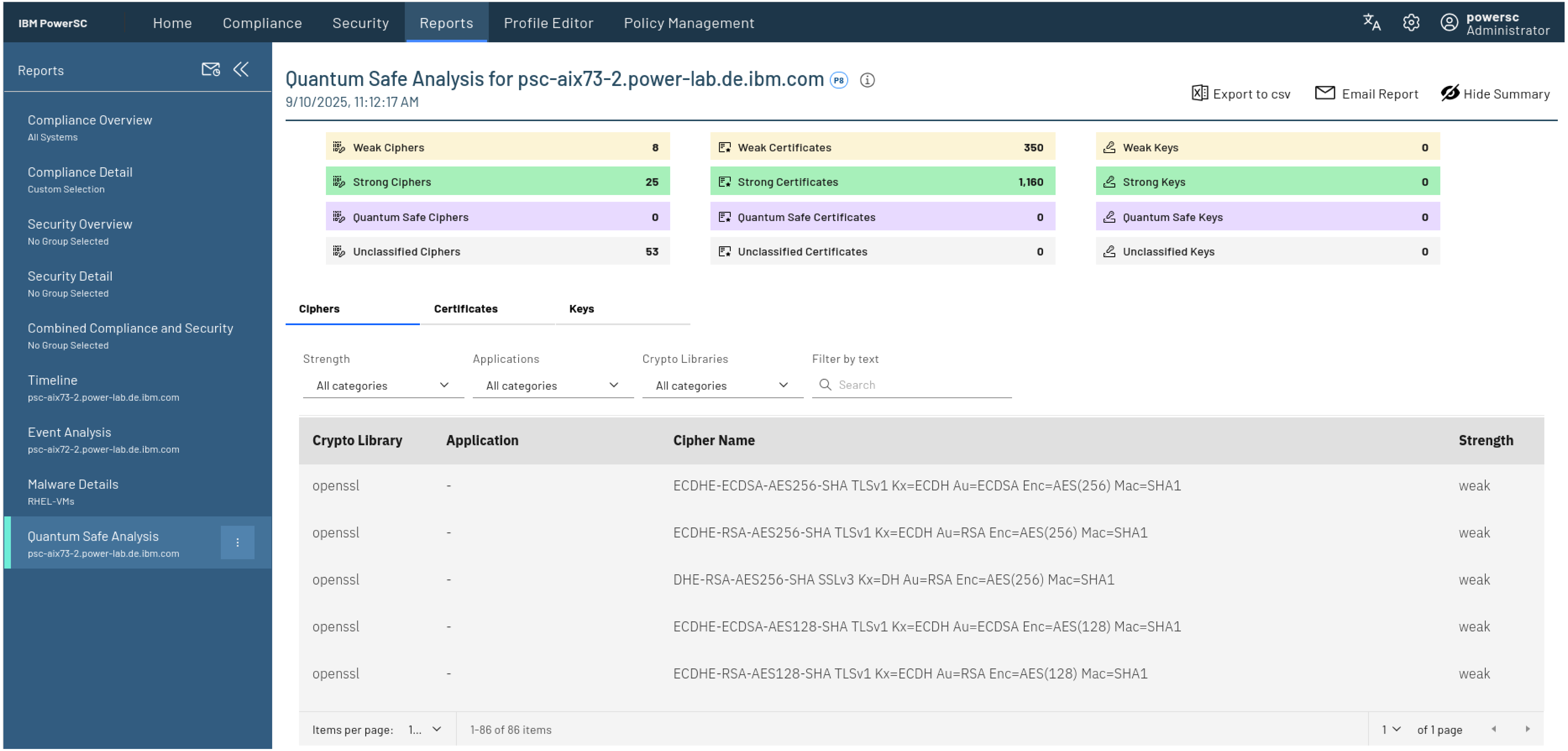
*…**an attacker** compromises the legacy signature via a quantum attack*

…the attacker would be able to obtain the symmetric encryption key and read all traffic

***However,** with Power11 LPM, the attacker cannot break the Quantum Safe signature, hence the symmetric encryption key is protected from Quantum attacks*



PowerSC – New Quantum Safe Inventory Report



IBM